ing mass and flavour states, and the most interesting parameter space corresponds to $|U_{\alpha 4}|^2 \ll 1$, i.e. ν_ℓ is mostly active and ν_4 is mostly sterile. The SN sterile neutrino production is discussed extensively in Ref. [15]. In summary, the most important production channels are the neutral current scattering on nucleons, $\nu_\alpha N \to \nu_s N$, and the charged current one, $\mu^- N \to \nu_s N$. Subleading processes are elastic $2 \to 2$ neutrino scatterings. A more detailed discussion on sterile neutrinos and their constraints arising from IA observations can be found in Ref. [24], where the importance of IA as probe of new physics was originally realized.

Dark photons

The DP is a $U(1)'$ gauge boson kinetically mixed with the SM photon [79–81]. In this context, the relevant terms in the DP Lagrangian are [80, 82]

$$\mathcal{L} = \frac{1}{2} m_{A'} A'_\mu A'^\mu - \frac{1}{4} F'_{\mu\nu} F'^{\mu\nu} - \frac{\epsilon}{2} F'_{\mu\nu} F^{\mu\nu} , \qquad (13)$$

where A' is the DP field, ϵ the mixing parameter, $F_{\mu\nu}$ the electromagnetic field strength tensor and $F'_{\mu\nu}$ the analogue for the DP. Being massive, DPs are provided with both transverse (T) and longitudinal (L) degrees of freedom. DP production in the SN core has been estimated in a series of papers [17, 83–87]. In particular, DPs are mainly emitted via proton bremsstrahlung $p\,N \to A\,p\,N$.

Following Ref. [27, 86], the number of DPs produced per unit volume and time in a SN is given by

$$\begin{aligned} \frac{dN^0_{A'}}{dV\,dt} &= \left.\frac{dN^0_{A'}}{dV\,dt}\right|_L + \left.\frac{dN^0_{A'}}{dV\,dt}\right|_T = \\ &= \int \frac{dE\,E^2 v}{2\,\pi^2}\, e^{-E/T} (\Gamma'_{\rm abs,L} + 2\Gamma'_{\rm abs,T}) , \end{aligned} \qquad (14)$$

where v is the velocity, $\Gamma'_{\rm abs,L/T}$ is the absorptive width of the DP for the longitudinal and transverse modes. The main absorption process in the SN core is inverse bremsstrahlung (ibr), thus we can write

$$\begin{aligned} \Gamma'_{\rm ibr,L/T} &= \frac{32}{3\pi} \frac{\alpha (\epsilon_m)^2_{\rm L/T} n_n n_p}{E^3} \left(\frac{\pi T}{m_N}\right)^{3/2} \\ &\times \langle \sigma^{(2)}_{np}(T) \rangle \left(\frac{m^2_{A'}}{E^2}\right)_{\rm L} , \end{aligned} \qquad (15)$$

with n_n and n_p the neutron and the proton number density, $m_N = 938$ MeV, $\langle \sigma^{(2)}_{np}(T) \rangle$ the averaged neutron-proton dipole scattering cross section from Ref. [88], $(\epsilon_m)^2_{\rm L/T}$ the in-medium mixing angle

$$(\epsilon_m)^2_{\rm L/T} = \frac{\epsilon^2}{(1 - {\rm Re}\Pi_{\rm L/T}/m^2_{A'})^2 + ({\rm Im}\,\Pi_{\rm L/T}/m^2_{A'})^2} , \qquad (16)$$

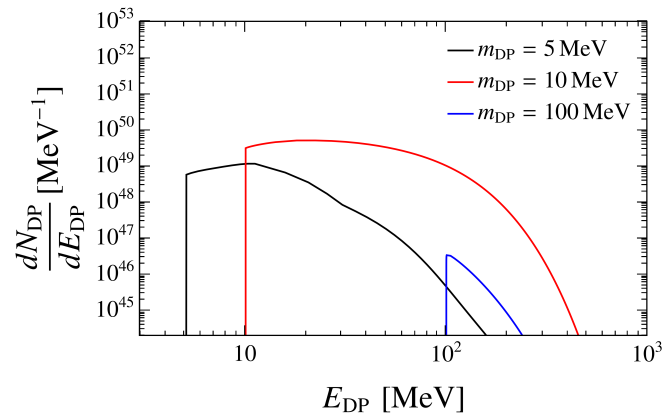

FIG. 3. DP production spectra via proton bremsstrahlung integrated over the SN volume and over the duration of the SN cooling phase. In this figure we set $\epsilon = 10^{-13}$.

where Π is the photon polarization tensor. The real part of the polarization tensor for the two modes is given by

$$\begin{aligned} {\rm Re}\,\Pi_{\rm L} &= \frac{3\omega_p^2}{v^2}(1-v^2)\left[\frac{1}{2v}\ln\left(\frac{1+v}{1-v}\right) - 1\right] , \\ {\rm Re}\,\Pi_{\rm T} &= \frac{3\omega_p^2}{2v^2}(1-v^2)\left[1 - \frac{1-v^2}{2v}\ln\left(\frac{1+v}{1-v}\right)\right] , \end{aligned} \qquad (17)$$

with ω_p the plasma frequency in the SN core. For typical SN conditions it can be estimated as

$$\omega_p^2 = \frac{4\pi\alpha_{\rm EM} n_e}{\sqrt{m_e^2 + (3\pi^2\, n_e)^{2/3}}} , \qquad (18)$$

where n_e is the electron number density.
The imaginary part of the polarization tensor can be written as

$${\rm Im}\,\Pi_{\rm L/T} = -E(1 - e^{-E/T})\Gamma_{\rm abs,L/T} , \qquad (19)$$

where $\Gamma_{\rm abs,L/T}$ is the absorptive width of SM photons

$$\Gamma'_{\rm ibr,L/T} = (\epsilon_m)^2_{\rm L/T}\Gamma_{\rm ibr,L/T} . \qquad (20)$$

Remarkably, in the interior of the SN core ${\rm Im}\,\Pi_{\rm L/T} \ll {\rm Re}\,\Pi_{\rm L/T}$. Therefore, Eq. (17) suggests that DP production is resonantly enhanced in the regions where ${\rm Re}\,\Pi_{\rm L/T} = m_{A'}$. This feature is directly encoded in the DP emission spectra reported in Fig. 3. We observe that DP production is efficient for masses $m_{\rm DP} \sim 15$ MeV, which is the common value assumed by the plasma frequency in the inner regions of the SN model considered in this work. Conversely, DP photon emission is sensibly suppressed for DP photon masses $m_{\rm DP} \gtrsim 50$ MeV, where resonant effects are washed out and Boltzmann suppression affects the production mechanism considered.